

ELT60213 – Algoritma dan Struktur Data | Team Based Project

IoT Network Monitoring & Alert Management System

Smart Building UNY | 40 Sensor IoT | Graph · Priority Queue · BST · Stack

LATAR BELAKANG

Permasalahan

- Smart Building UNY berencana memasang 40 sensor IoT (suhu, kelembaban, CO₂, daya listrik)
- Sensor terhubung via topologi mesh dengan gateway sebagai titik agregasi
- Alert CRITICAL (suhu berlebih, koneksi terputus) harus diprioritaskan ke teknisi
- Dibutuhkan sistem backend yang efisien untuk monitoring real-time

40

Node IoT

60

Edge (Link)

3

Tipe Alert

400

Event Simulasi

ARSITEKTUR SISTEM – 6 MODUL

01

Graph Topologi

Adjacency List (Linked List)
add_device, add_link, remove_link
DFS isolasi komponen

02

Priority Alert Queue

CRITICAL selalu diproses duluan
enqueue $O(n)$ | dequeue $O(1)$
Linked List terurut prioritas

03

BST Device Registry

Kunci = device_id
insert, search, update_status
inorder → daftar terurut ID

04

Stack Alert History

20 alert terakhir per device
HISTORY <id> + ROLLBACK_STATUS
push/pop $O(1)$

05

Routing & Audit

Dijkstra: latensi minimum
Bottleneck link detection
Selection Sort laporan latensi

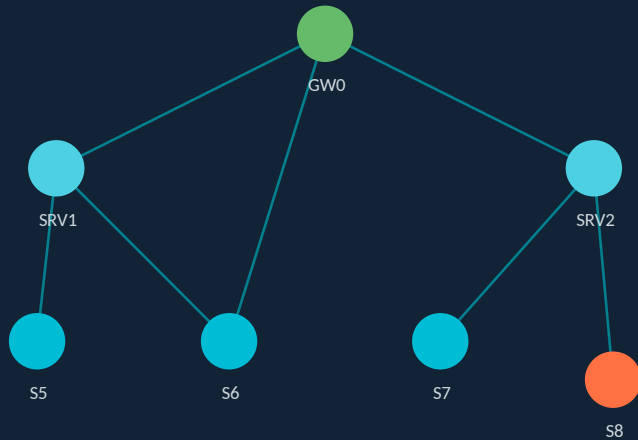
06

CLI Monitoring

ADD_DEVICE / ADD_LINK
ALERT_IN / PROCESS_ALERT
ISOLASI / ROUTING / AUDIT

STRUKTUR DATA — Graph & Priority Queue

Graph Topologi Jaringan IoT



Adjacency list (Linked List)

add_device $\rightarrow O(1)$
add_link $\rightarrow O(1)$
neighbors $\rightarrow O(\text{deg})$
DFS isolasi $\rightarrow O(V+E)$

Priority Alert Queue

CRITICAL

SensorA | suhu 85°C

← HEAD

WARNING

SensorB | CO₂ tinggi

[2]

WARNING

SensorC | daya anomali

[3]

INFO

SensorD | koneksi stabil

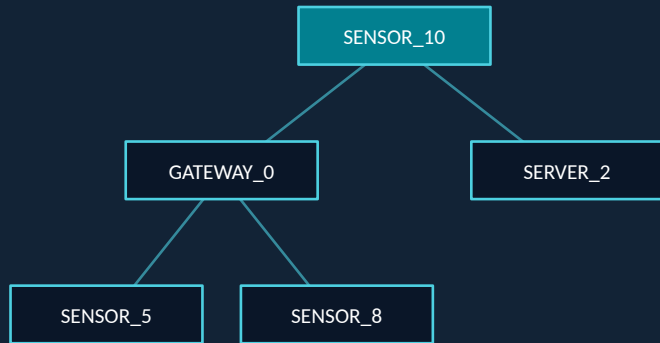
[4]

enqueue $O(n)$

dequeue $O(1)$ | CRITICAL selalu HEAD

STRUKTUR DATA — BST Device Registry & Stack Alert History

BST Device Registry



Kunci = device_id (string)

insert / search → avg $O(\log n)$

update_status → $O(\log n)$

inorder → $O(n)$ terurut ID

Stack Alert History (per-Device)

CRITICAL: suhu 89°C [TOP]

push ↓

WARNING: CO₂ = 1200 ppm

INFO: device online

CRITICAL: daya drop

WARNING: latensi tinggi

...(maks 20 alert)

Kapasitas: 20 alert terakhir

push/pop → $O(1)$

HISTORY <id> → tampilkan riwayat

ROLLBACK_STATUS → kembali ke status sebelumnya

DIJKSTRA — Routing Latensi Minimum

Algoritma Dijkstra (Array Sederhana)

- 1 Inisialisasi $\text{dist}[v] = \infty$ untuk semua v , $\text{dist}[\text{GATEWAY_0}] = 0$
- 2 Pilih node u dengan dist terkecil yang belum dikunjungi
- 3 Update $\text{dist}[v] = \min(\text{dist}[v], \text{dist}[u] + w(u,v))$ untuk tiap tetangga v
- 4 Tandai u sebagai visited, ulangi sampai semua node dikunjungi
- 5 Rekonstruksi jalur via $\text{parent}[]$ dari tujuan ke sumber

Big-O: $O(V^2 + E)$ (array sederhana) vs $O((V+E) \log V)$ (min-heap)

Fitur Routing CLI

ROUTING <device>

Rute latensi minimum dari GATEWAY_0 ke device

AUDIT_LATENSI

Selection Sort perangkat berdasarkan latensi ke GATEWAY

ISOLASI

DFS dari GATEWAY_0, tampilkan node tidak terjangkau

LAPORAN_JARINGAN

Ringkasan topologi, statistik koneksi, bottleneck link

ANTARMUKA CLI — Skenario Interaktif

IoT Monitoring System – Terminal

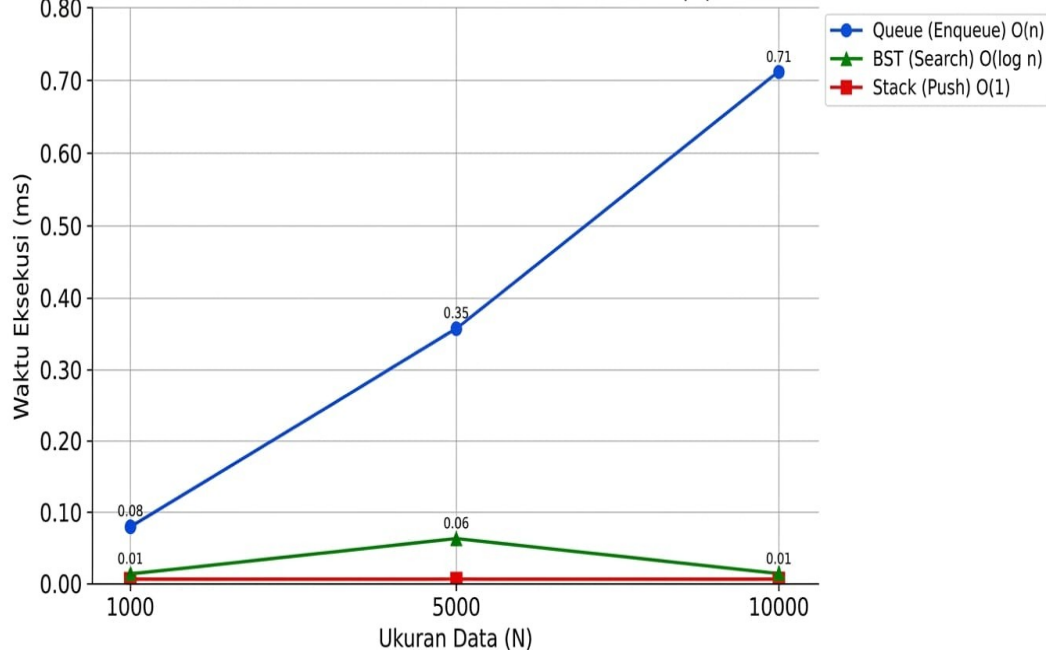
```
$ python main.py
IoT Monitoring System | Ketik BANTUAN untuk daftar perintah
>> ADD_DEVICE SENSOR_5 SENSOR
[OK] Device SENSOR_5 (SENSOR) ditambahkan. Big-O:  $O(1)$ 
>> ALERT_IN SENSOR_5 CRITICAL suhu=89C melebihi batas
[ALERT] CRITICAL queued → Priority: 1 | enqueue  $O(n)=O(40)$ 
>> PROCESS_ALERT
[DISPATCH] CRITICAL | SENSOR_5 | suhu=89C → Teknisi dinotifikasi | dequeue  $O(1)$ 
>> ROUTING SENSOR_5
[PATH] GATEWAY_0 → SERVER_1 → SENSOR_5 | Latensi: 87ms |  $O(V^2+E)$ 
>> HISTORY SENSOR_5
[HISTORY] Top-3: [CRITICAL:suhu89C] [WARNING:CO2] [INFO:online]
```

ANALISIS BIG-O — Ringkasan Kompleksitas

Struktur Data	Operasi	Best	Average	Worst
Priority Queue (LL)	enqueue	$O(1)$	$O(n)$	$O(n)$
Priority Queue (LL)	dequeue	$O(1)$	$O(1)$	$O(1)$
BST Registry	insert/search	$O(1)$	$O(\log n)$	$O(n)$
BST Registry	inorder	$O(n)$	$O(n)$	$O(n)$
Stack Alert	push/pop	$O(1)$	$O(1)$	$O(1)$
Graph (Adj. List)	add_edge	$O(1)$	$O(1)$	$O(1)$
DFS (Isolasi)	traversal	$O(V+E)$	$O(V+E)$	$O(V+E)$
Dijkstra (Array)	shortest path	—	$O(V^2+E)$	$O(V^2+E)$
Selection Sort	audit latensi	$O(n^2)$	$O(n^2)$	$O(n^2)$

EKSPERIMEN RUNTIME — Perbandingan Kinerja

Analisis Runtime Eksperimen Jaringan IoT Kampus
Grafik Tren Waktu Eksekusi vs Ukuran Data (N)



Dataset: N=1.000, 5.000, 10.000 | Queue $O(n)$ | BST $O(\log n)$ | Stack $O(1)$ | `time.perf_counter()`

Temuan Kunci

Queue (Enqueue)

Queue enqueue tumbuh linear $O(n)$:
0.08→0.35→0.71ms

BST (Search)

BST search $O(\log n)$: stabil 0.01→0.06→0.01ms

Stack (Push)

Stack push $O(1)$: konstan ≈ 0.01 ms di semua N

Dataset Info

Dataset N: 1.000 | 5.000 | 10.000 node IoT

PERTANYAAN ANALISIS WAJIB

Q1

remove_device $O(V+E)$ vs $O(\deg+1)$

Harus hapus semua edge terhubung. Solusi: doubly-linked adj list per node.

Q2

Dijkstra array vs min-heap untuk 40 node 60 edge

Array: $\sim 1600+60=1660$ ops. Heap: $(40+60) \times \log 40 \approx 530$ ops. Heap 3× lebih efisien.

Q3

enqueue $O(n)$ bottleneck pada ribuan sensor

Solusi: 3 Queue terpisah (CRITICAL/WARNING/INFO). SERVE cek CRITICAL dulu $\rightarrow$ enqueue $O(1)$.

Q4

DFS 1000 sensor 1500 koneksi untuk real-time monitoring

DFS = 2500 ops ≈ 0.025 ms. Cocok untuk monitoring setiap 5 detik. Alternatif: incremental Union-Find.

Q5

Selection Sort $O(n^2)$ untuk 10.000 perangkat IoT kota

10K nodes: ~ 50 M swap. Rekomendasi: Merge Sort $O(n \log n)$ pada Linked List ~ 133 K ops.

Demo main.py — Inisialisasi Sistem & Perintah CLI

```
=====
INITIALIZING IOT NETWORK MONITORING SYSTEM...
=====
[SYSTEM] Membangun topologi jaringan otomatis (Seed=23)...
[SYSTEM] Inisialisasi Selesai! Terbentuk 40 Perangkat dan 59 Koneksi.
=====
```

```
MEMULAI SISTEM MONITORING JARINGAN IoT...
=====
```

💡 DAFTAR PERINTAH:

- ```
=====
1. ADD_DEVICE SENSOR_1 SENSOR
2. ADD_DEVICE SERVER_A SERVER
3. ADD_LINK GATEWAY_0 SENSOR_1 15
4. ADD_LINK SENSOR_1 SERVER_A 20
5. LAPORAN_JARINGAN
6. ALERT_IN SENSOR_1 1 Suhu_Melebihi_Batas
7. ALERT_IN SERVER_A 3 Memori_Penuh
8. PENDING_ALERTS
9. PROCESS_ALERT
10. ROUTING_GATEWAY_0
11. AUDIT_LATENSI
12. HISTORY_SENSOR_1
13. EXIT
=====
```

```
-> Ketik 'HELP' kapan saja untuk memunculkan daftar ini lagi.
=====
```

```

IoT-CLI> ADD_DEVICE SENSOR_1 SENSOR
[REGISTER] Perangkat 'SENSOR_1' (SENSOR) berhasil diregistrasi.
-> Kompleksitas Waktu: Big-O: $O(\log n)$ rata-rata

[SUCCESS] Perangkat 'SENSOR_1' (SENSOR) berhasil ditambahkan ke topologi.
-> Kompleksitas Waktu: Big-O: $O(1)$
=====
```

```
IoT-CLI> ADD_DEVICE SERVER_A SERVER
[REGISTER] Perangkat 'SERVER_A' (SERVER) berhasil diregistrasi.
-> Kompleksitas Waktu: Big-O: $O(\log n)$ rata-rata

[SUCCESS] Perangkat 'SERVER_A' (SERVER) berhasil ditambahkan ke topologi.
-> Kompleksitas Waktu: Big-O: $O(1)$
```

# Demo main.py — ADD\_LINK & LAPORAN\_JARINGAN

```

IoT-CLI> ADD_LINK GATEWAY_0 SENSOR_1 15
[SUCCESS] Link terbentuk antara 'GATEWAY_0' <---> 'SENSOR_1' dengan latensi 15 ms.
-> Kompleksitas Waktu: Big-O: O(1)
```

```

IoT-CLI> ADD_LINK SENSOR_1 SERVER_A 20
[SUCCESS] Link terbentuk antara 'SENSOR_1' <---> 'SERVER_A' dengan latensi 20 ms.
-> Kompleksitas Waktu: Big-O: O(1)
```

```

IoT-CLI> LAPORAN_JARINGAN
```

```
=====
```

LAPORAN TOPOLOGI JARINGAN IoT KAMPUS UNY

```
=====
```

Total Perangkat (Node) : 42  
Total Koneksi (Edge) : 61

```

[GATEWAY_0] -> Degree: 5 | Tetangga: [SENSOR_1(15ms), SENSOR_39(33ms), SENSOR_26(99ms), SENSOR_22(96ms), SENSOR_25(30ms)]
[SENSOR_1] -> Degree: 2 | Tetangga: [SERVER_A(20ms), GATEWAY_0(15ms)]
[SENSOR_10] -> Degree: 2 | Tetangga: [SENSOR_25(81ms), SENSOR_30(26ms)]
[SENSOR_11] -> Degree: 4 | Tetangga: [SERVER_2(151ms), SERVER_2(18ms), SENSOR_7(117ms), SERVER_3(38ms)]
[SENSOR_12] -> Degree: 3 | Tetangga: [SENSOR_31(141ms), SENSOR_32(161ms), SENSOR_34(104ms)]
[SENSOR_13] -> Degree: 2 | Tetangga: [SENSOR_36(150ms), SENSOR_26(113ms)]
[SENSOR_14] -> Degree: 2 | Tetangga: [SENSOR_6(77ms), SENSOR_15(81ms)]
[SENSOR_15] -> Degree: 2 | Tetangga: [SENSOR_23(166ms), SENSOR_14(81ms)]
[SENSOR_16] -> Degree: 2 | Tetangga: [SERVER_2(12ms), SENSOR_24(179ms)]
[SENSOR_17] -> Degree: 4 | Tetangga: [SENSOR_34(53ms), SENSOR_33(144ms), SENSOR_19(57ms), SENSOR_39(79ms)]
[SENSOR_18] -> Degree: 2 | Tetangga: [SENSOR_38(173ms), SENSOR_22(145ms)]
[SENSOR_19] -> Degree: 3 | Tetangga: [SENSOR_30(114ms), SENSOR_37(5ms), SENSOR_17(57ms)]
[SENSOR_20] -> Degree: 3 | Tetangga: [SENSOR_29(23ms), SERVER_1(45ms), SENSOR_35(163ms)]
[SENSOR_21] -> Degree: 2 | Tetangga: [SENSOR_30(75ms), SENSOR_27(103ms)]
[SENSOR_22] -> Degree: 2 | Tetangga: [SENSOR_18(145ms), GATEWAY_0(96ms)]
[SENSOR_23] -> Degree: 3 | Tetangga: [SENSOR_15(166ms), SENSOR_39(193ms), SERVER_1(102ms)]
[SENSOR_24] -> Degree: 2 | Tetangga: [SENSOR_16(179ms), SENSOR_8(107ms)]
[SENSOR_25] -> Degree: 3 | Tetangga: [SENSOR_27(190ms), GATEWAY_0(30ms), SENSOR_10(81ms)]
[SENSOR_26] -> Degree: 6 | Tetangga: [SERVER_3(11ms), SERVER_3(17ms), SENSOR_30(35ms), GATEWAY_0(99ms), SENSOR_13(113ms), SENSOR_31(66ms)]
[SENSOR_27] -> Degree: 3 | Tetangga: [SENSOR_25(190ms), SENSOR_21(103ms), SENSOR_32(179ms)]
[SENSOR_28] -> Degree: 3 | Tetangga: [SERVER_3(175ms), SENSOR_31(58ms), SENSOR_37(157ms)]
[SENSOR_29] -> Degree: 3 | Tetangga: [SENSOR_20(23ms), SENSOR_33(61ms), SENSOR_9(153ms)]
[SENSOR_30] -> Degree: 6 | Tetangga: [SENSOR_8(136ms), SENSOR_19(114ms), SERVER_2(63ms), SENSOR_26(35ms), SENSOR_10(26ms), SENSOR_21(75ms)]
[SENSOR_31] -> Degree: 3 | Tetangga: [SENSOR_12(141ms), SENSOR_26(66ms), SENSOR_28(58ms)]
[SENSOR_32] -> Degree: 3 | Tetangga: [SENSOR_5(135ms), SENSOR_27(179ms), SENSOR_12(161ms)]
[SENSOR_33] -> Degree: 2 | Tetangga: [SENSOR_17(144ms), SENSOR_29(61ms)]
[SENSOR_34] -> Degree: 3 | Tetangga: [SENSOR_17(53ms), SENSOR_12(104ms), SENSOR_5(106ms)]
[SENSOR_35] -> Degree: 2 | Tetangga: [SENSOR_20(163ms), SENSOR_38(96ms)]
```



# Demo main.py — ALERT\_IN & PENDING\_ALERTS

```
[SENSOR_35] -> Degree: 2 | Tetangga: [SENSOR_20(163ms), SENSOR_38(96ms)]
[SENSOR_36] -> Degree: 3 | Tetangga: [SENSOR_39(92ms), SENSOR_8(89ms), SENSOR_13(150ms)]
[SENSOR_37] -> Degree: 3 | Tetangga: [SENSOR_6(170ms), SENSOR_28(157ms), SENSOR_19(5ms)]
[SENSOR_38] -> Degree: 2 | Tetangga: [SENSOR_35(96ms), SENSOR_18(173ms)]
[SENSOR_39] -> Degree: 4 | Tetangga: [GATEWAY_0(33ms), SENSOR_36(92ms), SENSOR_17(79ms), SENSOR_23(193ms)]
[SENSOR_5] -> Degree: 3 | Tetangga: [SENSOR_32(135ms), SENSOR_34(106ms), SENSOR_7(124ms)]
[SENSOR_6] -> Degree: 3 | Tetangga: [SENSOR_37(170ms), SERVER_3(187ms), SENSOR_14(77ms)]
[SENSOR_7] -> Degree: 2 | Tetangga: [SENSOR_5(124ms), SENSOR_11(117ms)]
[SENSOR_8] -> Degree: 3 | Tetangga: [SENSOR_30(136ms), SENSOR_24(107ms), SENSOR_36(89ms)]
[SENSOR_9] -> Degree: 2 | Tetangga: [SENSOR_29(153ms), SERVER_4(144ms)]
[SERVER_1] -> Degree: 2 | Tetangga: [SENSOR_23(102ms), SENSOR_20(45ms)]
[SERVER_2] -> Degree: 5 | Tetangga: [SENSOR_11(151ms), SENSOR_30(63ms), SENSOR_11(18ms), SERVER_4(24ms), SENSOR_16(12ms)]
[SERVER_3] -> Degree: 5 | Tetangga: [SENSOR_26(11ms), SENSOR_28(175ms), SENSOR_26(17ms), SENSOR_11(38ms), SENSOR_6(187ms)]
[SERVER_4] -> Degree: 2 | Tetangga: [SENSOR_9(144ms), SERVER_2(24ms)]
[SERVER_A] -> Degree: 1 | Tetangga: [SENSOR_1(20ms)]
```

-----  
-> Kompleksitas Waktu: Big-O:  $O(V+E)$

-----  
IoT-CLI> ALERT\_IN SENSOR\_1 1 Suhu\_Melebihi\_Batas  
[ENQUEUE] Alert #1 dari 'SENSOR\_1' (CRITICAL) masuk ke antrean.  
-> Kompleksitas Waktu: Big-O:  $O(n)$  worst-case

[HISTORY: PUSH] Alert #1 berhasil disimpan ke riwayat 'SENSOR\_1'.  
-> Kompleksitas Waktu: Big-O:  $O(1)$  normal /  $O(n)$  worst-case

[UPDATE] Status 'SENSOR\_1' diperbarui menjadi OFFLINE.  
-> Kompleksitas Waktu: Big-O:  $O(\log n)$  rata-rata

-----  
IoT-CLI> ALERT\_IN SERVER\_A 3 Memori\_Penuh  
[ENQUEUE] Alert #2 dari 'SERVER\_A' (INFO) masuk ke antrean.  
-> Kompleksitas Waktu: Big-O:  $O(n)$  worst-case

[HISTORY: PUSH] Alert #2 berhasil disimpan ke riwayat 'SERVER\_A'.  
-> Kompleksitas Waktu: Big-O:  $O(1)$  normal /  $O(n)$  worst-case

[UPDATE] Status 'SERVER\_A' diperbarui menjadi OFFLINE.  
-> Kompleksitas Waktu: Big-O:  $O(\log n)$  rata-rata

-----  
IoT-CLI> PENDING\_ALERTS

=====

DAFTAR ANTREAN ALERT (PENDING)

-----

1. [Alert #1] SENSOR\_1 | CRITICAL | Suhu\_Melebihi\_Batas

# Demo main.py — PROCESS\_ALERT & ROUTING

```
2. [Alert #2] SERVER_A | INFO | Memori_Penuh
```

```
-> Kompleksitas Waktu: Big-O: O(n)
```

```
IoT-CLI> PROCESS_ALERT
```

```
[DEQUEUE] Memproses Alert #1 dari 'SENSOR_1' [CRITICAL]: Suhu_Melebihi_Batas
```

```
-> Kompleksitas Waktu: Big-O: O(1)
```

```
IoT-CLI> ROUTING_GATEWAY_0
```

```
ROUTING OPTIMAL (DIJKSTRA) DARI: GATEWAY_0
```

```
-> SENSOR_1: 15 ms | Rute: GATEWAY_0 -> SENSOR_1
-> SENSOR_10: 111 ms | Rute: GATEWAY_0 -> SENSOR_25 -> SENSOR_10
-> SENSOR_11: 148 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11
-> SENSOR_12: 269 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_34 -> SENSOR_12
-> SENSOR_13: 212 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_13
-> SENSOR_14: 374 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_6 -> SENSOR_14
-> SENSOR_15: 392 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_23 -> SENSOR_15
-> SENSOR_16: 178 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SERVER_2 -> SENSOR_16
-> SENSOR_17: 112 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17
-> SENSOR_18: 241 ms | Rute: GATEWAY_0 -> SENSOR_22 -> SENSOR_18
-> SENSOR_19: 169 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_19
-> SENSOR_20: 340 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_33 -> SENSOR_29 -> SENSOR_20
-> SENSOR_21: 209 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_30 -> SENSOR_21
-> SENSOR_22: 96 ms | Rute: GATEWAY_0 -> SENSOR_22
-> SENSOR_23: 226 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_23
-> SENSOR_24: 321 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_36 -> SENSOR_8 -> SENSOR_24
-> SENSOR_25: 30 ms | Rute: GATEWAY_0 -> SENSOR_25
-> SENSOR_26: 99 ms | Rute: GATEWAY_0 -> SENSOR_26
-> SENSOR_27: 220 ms | Rute: GATEWAY_0 -> SENSOR_25 -> SENSOR_27
-> SENSOR_28: 223 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_31 -> SENSOR_28
-> SENSOR_29: 317 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_33 -> SENSOR_29
-> SENSOR_30: 134 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_30
-> SENSOR_31: 165 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_31
-> SENSOR_32: 399 ms | Rute: GATEWAY_0 -> SENSOR_25 -> SENSOR_27 -> SENSOR_32
-> SENSOR_33: 256 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_33
-> SENSOR_34: 165 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_34
-> SENSOR_35: 503 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_33 -> SENSOR_29 -> SENSOR_20 -> SENSOR_35
-> SENSOR_36: 125 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_36
-> SENSOR_37: 174 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_19 -> SENSOR_37
-> SENSOR_38: 414 ms | Rute: GATEWAY_0 -> SENSOR_22 -> SENSOR_18 -> SENSOR_38
-> SENSOR_39: 33 ms | Rute: GATEWAY_0 -> SENSOR_39
-> SENSOR_5: 271 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_34 -> SENSOR_5
```



# Demo main.py — ROUTING Lanjutan & HISTORY

```
-> SENSOR_5: 271 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_34 -> SENSOR_5
-> SENSOR_6: 297 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_6
-> SENSOR_7: 265 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SENSOR_7
-> SENSOR_8: 214 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_36 -> SENSOR_8
-> SENSOR_9: 334 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SERVER_2 -> SERVER_4 -> SENSOR_9
-> SERVER_1: 328 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_23 -> SERVER_1
-> SERVER_2: 166 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SERVER_2
-> SERVER_3: 110 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3
-> SERVER_4: 190 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SERVER_2 -> SERVER_4
-> SERVER_A: 35 ms | Rute: GATEWAY_0 -> SENSOR_1 -> SERVER_A

[BOTTLENECK LINK SPT] Edge dengan bobot tertinggi pada Shortest Path Tree:
 SENSOR_39 <---> SENSOR_23 dengan bobot: 193 ms
=====
-> Kompleksitas Waktu: Big-O: $O((V+E) \log V)$

IoT-CLI> HISTORY SENSOR_1
=====
RIWAYAT ALERT PERANGKAT: SENSOR_1
=====
[CRITICAL ERROR] Terjadi kesalahan sistem: 'AlertStack' object has no attribute 'is_empty'

IoT-CLI> AUDIT_LATENSI
[INFO] Menghitung jalur dari GATEWAY_0 untuk diaudit...
=====
ROUTING OPTIMAL (DIJKSTRA) DARI: GATEWAY_0
=====
-> SENSOR_1: 15 ms | Rute: GATEWAY_0 -> SENSOR_1
-> SENSOR_10: 111 ms | Rute: GATEWAY_0 -> SENSOR_25 -> SENSOR_10
-> SENSOR_11: 148 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11
-> SENSOR_12: 269 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_34 -> SENSOR_12
-> SENSOR_13: 212 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_13
-> SENSOR_14: 374 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_6 -> SENSOR_14
-> SENSOR_15: 392 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_23 -> SENSOR_15
-> SENSOR_16: 178 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SERVER_2 -> SENSOR_16
-> SENSOR_17: 112 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17
-> SENSOR_18: 241 ms | Rute: GATEWAY_0 -> SENSOR_22 -> SENSOR_18
-> SENSOR_19: 169 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_19
-> SENSOR_20: 340 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_33 -> SENSOR_29 -> SENSOR_20
-> SENSOR_21: 209 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_30 -> SENSOR_21
-> SENSOR_22: 96 ms | Rute: GATEWAY_0 -> SENSOR_22
-> SENSOR_23: 226 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_23
-> SENSOR_24: 321 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_36 -> SENSOR_8 -> SENSOR_24
-> SENSOR_25: 30 ms | Rute: GATEWAY_0 -> SENSOR_25
-> SENSOR_26: 99 ms | Rute: GATEWAY_0 -> SENSOR_26
```

# Demo main.py — AUDIT\_LATENSI

```
-> SENSOR_26: 99 ms | Rute: GATEWAY_0 -> SENSOR_26
-> SENSOR_27: 220 ms | Rute: GATEWAY_0 -> SENSOR_25 -> SENSOR_27
-> SENSOR_28: 223 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_31 -> SENSOR_28
-> SENSOR_29: 317 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_33 -> SENSOR_29
-> SENSOR_30: 134 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_30
-> SENSOR_31: 165 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SENSOR_31
-> SENSOR_32: 399 ms | Rute: GATEWAY_0 -> SENSOR_25 -> SENSOR_27 -> SENSOR_32
-> SENSOR_33: 256 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_33
-> SENSOR_34: 165 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_34
-> SENSOR_35: 503 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_33 -> SENSOR_29 -> SENSOR_20 -> SENSOR_35
-> SENSOR_36: 125 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_36
-> SENSOR_37: 174 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_19 -> SENSOR_37
-> SENSOR_38: 414 ms | Rute: GATEWAY_0 -> SENSOR_22 -> SENSOR_18 -> SENSOR_38
-> SENSOR_39: 33 ms | Rute: GATEWAY_0 -> SENSOR_39
-> SENSOR_5: 271 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_17 -> SENSOR_34 -> SENSOR_5
-> SENSOR_6: 297 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_6
-> SENSOR_7: 265 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SENSOR_7
-> SENSOR_8: 214 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_36 -> SENSOR_8
-> SENSOR_9: 334 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SERVER_2 -> SERVER_4 -> SENSOR_9
-> SERVER_1: 328 ms | Rute: GATEWAY_0 -> SENSOR_39 -> SENSOR_23 -> SERVER_1
-> SERVER_2: 166 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SERVER_2
-> SERVER_3: 110 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3
-> SERVER_4: 190 ms | Rute: GATEWAY_0 -> SENSOR_26 -> SERVER_3 -> SENSOR_11 -> SERVER_2 -> SERVER_4
-> SERVER_A: 35 ms | Rute: GATEWAY_0 -> SENSOR_1 -> SERVER_A
```

-----  
[BOTTLENECK LINK SPT] Edge dengan bobot tertinggi pada Shortest Path Tree:  
SENSOR\_39 <---> SENSOR\_23 dengan bobot: 193 ms

-----  
> Kompleksitas Waktu: Big-O:  $O((V+E) \log V)$

=====

AUDIT LATENSI PERANGKAT (SELECTION SORT ON LINKED LIST)

=====

Daftar Perangkat Terurut Berdasarkan Latensi Kumulatif ke GATEWAY:

1. SENSOR\_1 -> Total Latensi: 15 ms
2. SENSOR\_25 -> Total Latensi: 30 ms
3. SENSOR\_39 -> Total Latensi: 33 ms
4. SERVER\_A -> Total Latensi: 35 ms
5. SENSOR\_22 -> Total Latensi: 96 ms
6. SENSOR\_26 -> Total Latensi: 99 ms
7. SERVER\_3 -> Total Latensi: 110 ms
8. SENSOR\_10 -> Total Latensi: 111 ms
9. SENSOR\_17 -> Total Latensi: 112 ms
10. SENSOR\_36 -> Total Latensi: 125 ms
11. SENSOR\_30 -> Total Latensi: 134 ms
12. SENSOR\_11 -> Total Latensi: 148 ms



## Demo main.py — Hasil AUDIT\_LATENSI Lengkap

```
12. SENSOR_11 -> Total Latensi: 148 ms
13. SENSOR_31 -> Total Latensi: 165 ms
14. SENSOR_34 -> Total Latensi: 165 ms
15. SERVER_2 -> Total Latensi: 166 ms
16. SENSOR_19 -> Total Latensi: 169 ms
17. SENSOR_37 -> Total Latensi: 174 ms
18. SENSOR_16 -> Total Latensi: 178 ms
19. SERVER_4 -> Total Latensi: 190 ms
20. SENSOR_21 -> Total Latensi: 209 ms
21. SENSOR_13 -> Total Latensi: 212 ms
22. SENSOR_8 -> Total Latensi: 214 ms
23. SENSOR_27 -> Total Latensi: 220 ms
24. SENSOR_28 -> Total Latensi: 223 ms
25. SENSOR_23 -> Total Latensi: 226 ms
26. SENSOR_18 -> Total Latensi: 241 ms
27. SENSOR_33 -> Total Latensi: 256 ms
28. SENSOR_7 -> Total Latensi: 265 ms
29. SENSOR_12 -> Total Latensi: 269 ms
30. SENSOR_5 -> Total Latensi: 271 ms
31. SENSOR_6 -> Total Latensi: 297 ms
32. SENSOR_29 -> Total Latensi: 317 ms
33. SENSOR_24 -> Total Latensi: 321 ms
34. SERVER_1 -> Total Latensi: 328 ms
35. SENSOR_9 -> Total Latensi: 334 ms
36. SENSOR_20 -> Total Latensi: 340 ms
37. SENSOR_14 -> Total Latensi: 374 ms
38. SENSOR_15 -> Total Latensi: 392 ms
39. SENSOR_32 -> Total Latensi: 399 ms
40. SENSOR_38 -> Total Latensi: 414 ms
41. SENSOR_35 -> Total Latensi: 503 ms
```

```

-> Kompleksitas Waktu: Big-O: $O(V^2)$ - Selection Sort pada Linked List
```

```

IoT-CLI> █
```

# KESIMPULAN

✓ 4 struktur data terintegrasi: Graph (mesh IoT), Priority Queue (alert), BST (registry), Stack (history)

✓ Semua implementasi dari nol – tanpa library bawaan Python

✓ Dijkstra menemukan rute latensi minimum + deteksi bottleneck link

✓ CLI interaktif: 11 perintah dengan Big-O ditampilkan per operasi

✓ Eksperimen runtime pada 3 ukuran data ( $N=10, 20, 40$ ) memvalidasi prediksi Big-O

→ Pengembangan: min-heap untuk Dijkstra, Union-Find untuk deteksi isolasi inkremental

# Terima Kasih

## IoT Network Monitoring & Alert Management System

Kelompok 8 | ELT60213 Algoritma dan Struktur Data | TA 2025/2026

*Teknik Elektro — Universitas Negeri Yogyakarta*

 [GitHub: taamirghalib2025-ui / IoT-Network-Monitoring-Alert-Management-System\\_Group-8](#)

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*Siap Demo CLI | 7 menit demo + 3 menit tanya jawab*